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Dynamic Calibration of Saturation in Reservoir Simulation Initialization

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Abstract

Accurate initialization of water saturation is a critical step in reservoir simulation, particularly in heterogeneous carbonate reservoirs where dynamic parameters such as capillary pressure – water saturation functions are often burdened with high uncertainty. This study introduces a novel, data-driven workflow to calibrate dynamic saturation initialization by optimizing these uncertain parameters using a machine learning (ML)-based proxy model. The workflow aligns simulation-initialized saturation with well-log-derived saturation through systematic variation of capillary pressure endpoints, water-oil contact, and rock typing scenarios, aiming to improve initialization quality and simulation performance.

The proposed approach was implemented on an open-source carbonate reservoir model composed of 144 geological realizations, grouped into three distinct rock typing scenarios. In this study, three representative realizations corresponding to the 3, 5, and 7 rock type configurations were selected to evaluate which rock typing scheme best represents the reservoir conditions. For each rock typing scenario, 200 simulation cases were generated by varying capillary pressure endpoints and water-oil contact depths. The saturation mismatch between simulation output and well-log-derived saturation was computed for each case. Machine learning regression models were trained using these outputs and inputs to predict saturation mismatch. Multiple ML algorithms were evaluated, and random forest regression provided the best performance. Using the trained model, the dynamic parameters were then optimized to minimize the predicted saturation mismatch, enabling more accurate and efficient initialization.

Two calibration approaches were investigated: Method 1, which predicts and optimizes the average saturation mismatch per simulation case, and Method 2, which predicts and optimizes saturation mismatch at each depth point. The results indicate that Method 2 provides superior calibration performance, with the five-rock-type model yielding the lowest saturation mismatch after optimization. The random forest model prediction of this model achieved a high predictive performance, with an R^2 score of 0.933 in estimating saturation mismatch across the simulation dataset. The optimization process led to a reduction in average saturation mismatch by more than 50%, improving from 0.34 to 0.17. As a result, the calibrated initialization contributed to more stable numerical performance in the reservoir simulator, reducing simulation runtime by approximately 52%. Furthermore, the improved saturation distribution facilitated faster and more reliable history matching. This case study demonstrates that integrating machine learning into the dynamic

initialization process can significantly reduce uncertainty, improve model representativeness, and increase efficiency across the reservoir modeling workflow.

This study presents a novel integration of machine learning and dynamic parameter optimization to calibrate reservoir simulation initialization. It provides a systematic, automated alternative to manual trial-and-error, reducing effort while improving model quality and stability. Additionally, the approach provides insight into which rock typing configuration best represents the reservoir, thereby enhancing uncertainty quantification and supporting more informed decision-making in reservoir development planning.

Introduction

In reservoir simulation, initialization is the process to establish the initial condition of the reservoir, such as pressure, fluid saturation, and rock property distribution, before starting the fluid flow simulation calculation. This step is critical for accurately estimating hydrocarbon reserves and for developing dynamic models that realistically represent reservoir behavior. The overall accuracy of a reservoir simulation heavily depends on the quality of this initialization, which in turn is governed by the robustness of the reservoir model (Worthington and Hattingh, 2012). Considering that initialization should represent the reservoir's equilibrium condition, saturation is one of the most complex properties to model, as it is highly sensitive to static factors such as capillary pressure, and fluid contacts.

There are several approaches to initializing water saturation in a dynamic reservoir model. One method involves directly importing the initial water saturation from a static model. While this approach is straightforward and can yield consistency in fluid volumes between static and dynamic models, it often fails to ensure that the cells are correctly equilibrated under capillary and gravitational forces. As a result, it may reduce the model's capability to capture the dynamic behavior of transition zones, especially if saturation-height relationships based on capillary pressure data are not properly applied (Worthington and Hattingh, 2012).

There is also J-Function approach that represents the relationship between water saturation and capillary pressure through a dimensionless function that normalizes rock properties, enabling better capture of the variability across multiple rock types. In contrast, a more widely applied method in reservoir simulation determines water saturation by solving the capillary–gravity equilibration, which balances capillary and gravitational forces to define the initial fluid distribution within the reservoir (Kachuma, Hild, and Belushko, 2021). Using this method, water saturation S_w is computed by solving a one-dimensional ordinary differential equation (ODE) that reflects this equilibrium:

$$\frac{\partial P_c(S_w)}{\partial h} = -g(\rho_o - \rho_w)$$

where P_c is capillary pressure as a function of the water saturation, representing the pressure difference between oil and water phase. This pressure difference is directly related to the hydrostatic pressure from the density difference between oil and water ($\rho_o - \rho_w$) along with the gravitational constant g and the height h which measured upward from the free water level.

However, the challenge in this method is that saturation initialized in the reservoir may differ than the saturation derived from the well log. This mismatch between log-derived and capillary pressure-derived water saturation during initialization will increase the model uncertainty and may complicate the history matching process, often requiring additional time to align the model with observed reservoir data (Al-Jenaibi, Hammadi, and Salameh, 2008). Therefore, careful evaluation and validation of dynamic saturation initialization should be conducted as a critical step before proceeding with the simulation study.

This study aims to address this challenge by introducing a data-driven workflow for calibrating dynamic saturation initialization by optimizing key uncertain parameters through a machine learning (ML)-based proxy model. The objective is to minimize the discrepancy between simulation-initialized saturation and well-log-derived saturation by dynamically calibrating the capillary pressure – water saturation function,

and oil water contact (OWC), eventually enhancing the accuracy of reservoir initialization and improving overall simulation performance.

Field Description

This study focuses on the challenges of saturation initialization in heterogeneous carbonate reservoirs, considering that dynamic parameters such as capillary pressure – water saturation functions are often associated with high uncertainty. An open-source carbonate reservoir model called COSTA which was published in 2022 shows sufficient data to provide for this study. The COSTA model offers a comprehensive dataset that reflects a wide spectrum of geological, petrophysical, and geomodelling uncertainties commonly encountered in carbonate reservoirs. These uncertainties were addressed by generating multiple geological scenarios and constructing various reservoir architectures, each exhibiting distinct dynamic behaviors (Costa Gomes, Geiger, and Arnold, 2021).

The COSTA model includes 144 realizations that represent a range of geological and modeling uncertainties. These realizations are constructed using variations in the stratigraphic framework, data sources for porosity and facies modeling, and different anisotropic variogram ranges. This ensemble captures the inherent complexity of carbonate reservoirs, particularly in terms of facies heterogeneity and distribution. As part of the modeling process, the facies were categorized into 3, 5, and 7 reservoir rock type scenarios, each consisting of 48 realizations. Since some realizations varied only in independent properties such as porosity and permeability, which do not affect the capillary pressure–saturation function, it was decided to focus on three representative realizations corresponding to the 3, 5, and 7 reservoir rock type configurations. Each realization was generated using core data for porosity modeling and applied arithmetic averaging for property distribution, as illustrated in Figure 1.

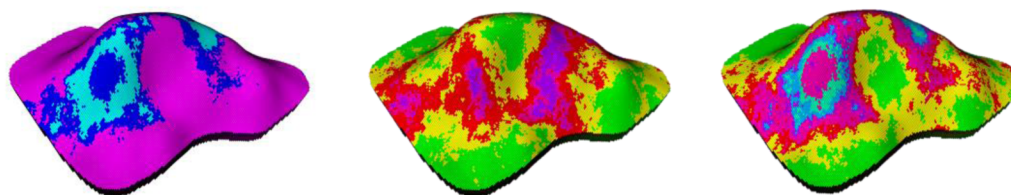


Figure 1—The 3D visualization of COSTA model based on its rock type scenario (a) The model on the left represents COSTA model with 3 rock typing; (b) The model in the middle represents COSTA model with 5 rock typing; and (c) The model on the right represents COSTA model with 7 rock typing.

One of the key reasons for selecting the COSTA model in this study is its extensive, well log dataset. The model includes 15 well-log data with high-quality saturation measurements, providing a robust basis for evaluating saturation initialization in reservoir simulation. This comprehensive dataset allows for direct comparison between saturation from simulation initialization and saturation derived from well logs, enabling a more reliable assessment of the initialization accuracy.

Uncertain Parameters in Modelling Saturation

As previously discussed, the mismatch in water saturation between reservoir simulation initialization and well log measurements arise from the use of dynamic parameters in the simulation, which are employed to solve capillary-gravity equilibration. These parameters are derived through various methods that inherently involve uncertainties related to data acquisition and interpretation. Consequently, these uncertainties contribute to discrepancies between simulated and measured water saturation values.

Capillary Pressure – Water Saturation Functions

In reservoir simulation, the capillary pressure – water saturation relationship is a critical input typically provided as a tabular data, which is then used to calculate water saturation in each grid cell based on the solution of capillary-gravity equilibrium. This function is generally derived from laboratory measurements and subsequently analyzed to produce the capillary pressure curve, as illustrated in Figure 2. However, accurately defining this function is challenging due to significant uncertainties arising from variations in rock and fluid properties, as well as complex pore-scale phenomena and hysteresis effects that cause different fluid behaviors during drainage and imbibition (Zhou et al., 2017).

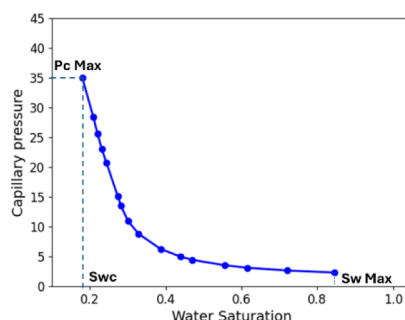


Figure 2—The capillary pressure – water saturation function with its three endpoints of a rock type: (a) Maximum capillary pressure (Pc Max); (b) Connate water saturation (Swc); (c) Maximum water saturation (Sw Max).

To account for these uncertainties, the capillary pressure – water saturation function can be calibrated to achieve a better match between simulation and observed saturation. This is done without altering the physical trend of the rock properties by adjusting the endpoints of the function. There are three key endpoints, as shown in Figure 2: the maximum capillary pressure (Pc Max), connate water saturation (Swc), and maximum water saturation (Sw Max). These parameters were systematically varied to generate simulation data that can be used to train the machine learning proxy model.

In this study, the COSTA model with 3, 5, and 7 rock type configurations was selected to evaluate the impact of geological heterogeneity on saturation initialization. Each rock type configuration corresponds to a distinct number of capillary pressure – water saturation functions, meaning that a model with 5 rock types includes 5 unique capillary pressure curves. These functions were derived from a comprehensive SCAL database developed by the authors of COSTA model, which contains 110 carbonate plug samples characterized by mercury injection capillary pressure (MICP) and pore throat size distribution measurements. Each sample is also linked to porosity, permeability, and Winland R35 values to ensure consistency in rock typing and accurate representation of capillary pressure behavior (Costa Gomes, Geiger, and Arnold, 2021).

Oil Water Contact

The oil-water contact (OWC) serves as a reference depth for calculating capillary pressure in each simulation grid cell. OWC is typically determined from water saturation profiles derived from resistivity logs, but this method is often uncertain due to poor downhole conditions and ambiguous log interpretations. Despite these uncertainties and variations between wells caused by reservoir heterogeneity, an average OWC is commonly used in reservoir simulation and reserve estimation (Al-Rubaye et al., 2020).

In the COSTA model, uncertainties within the ensemble were addressed by varying both the number of OWC and the thicknesses of capillary transition zones (Costa Gomes, Geiger, and Arnold, 2021). The assigned OWC values varied depending on the stratigraphic framework applied in each scenario. In this study, the OWC defined by the original model was used as a base case, which was then systematically adjusted to generate simulation data for input into the machine learning model.

Dynamic Calibration Workflow

The dynamic calibration workflow is designed to systematically adjust uncertain reservoir parameters to improve the accuracy of water saturation initialization. This process integrates machine learning as proxy model with optimization techniques to minimize the saturation mismatch.

Saturation Mismatch Calculation

Based on the original COSTA model, a significant discrepancy is observed between the simulated water saturation that is based on measured capillary pressure data and the measurements from well logs, as illustrated in Figure 3. When the initial saturation deviates substantially from log-derived values, it can negatively impact the reservoir simulation workflow, particularly during the history matching process. To assess this mismatch, the saturation difference must be quantified at each depth by comparing simulation output with well log measurements.

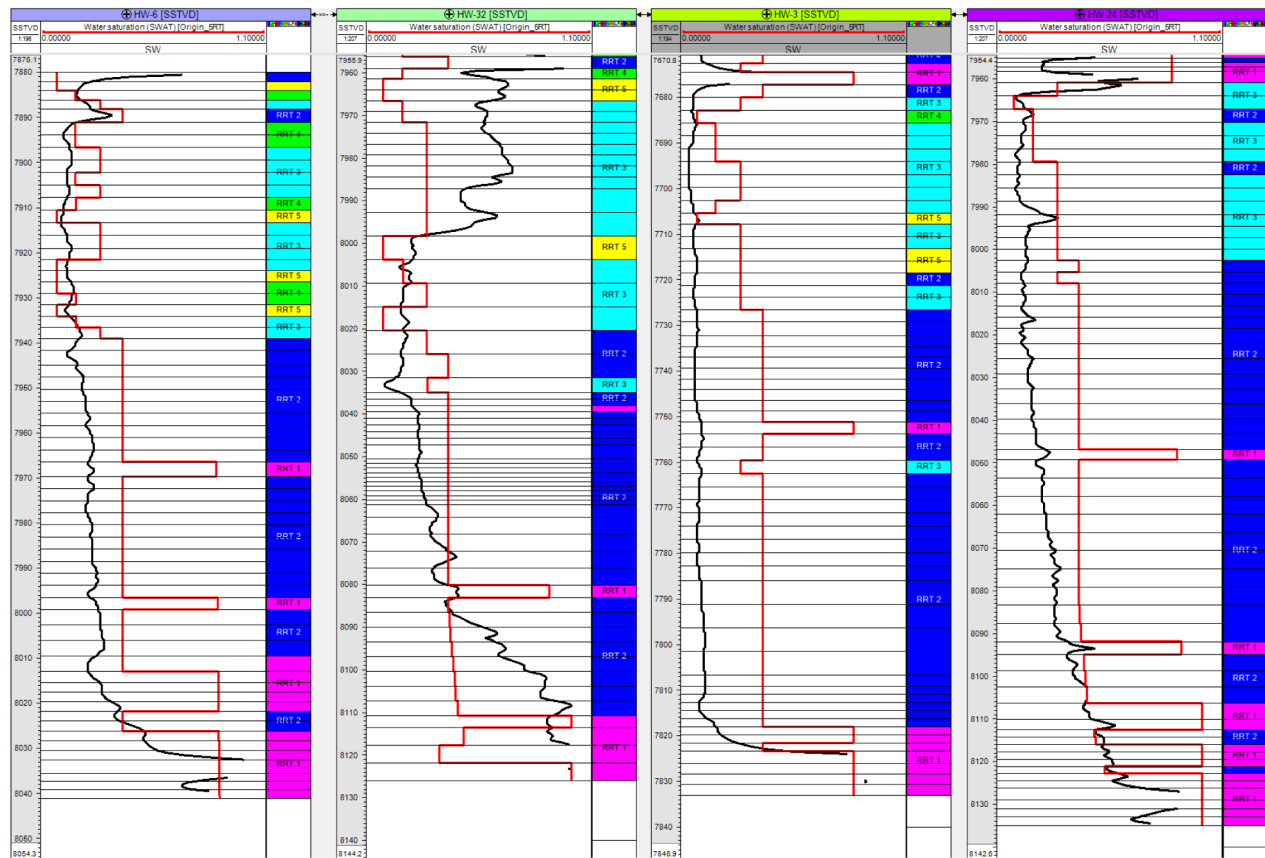


Figure 3—An example of water saturation difference between reservoir simulation initialization and well log measurement of a COSTA Model with 5 rock types. The black line is water saturation measured in the well log, and the red line is the saturation from reservoir simulation initialization.

To align the simulation results with well log data, it is necessary to interpolate water saturation values at the same depth intervals recorded in the logs. This is particularly important because simulation grid blocks and well log measurements often have different depth resolutions. The following linear interpolation formula is used to estimate water saturation at any given depth between two known data points:

$$S_{w,interpolated} = S_{w1} + \frac{S_{w2} - S_{w1}}{Depth_2 - Depth_1} \times (Depth - Depth_1)$$

In this equation, S_{w1} and S_{w2} are the water saturation values at known depth $Depth_1$ and $Depth_2$, respectively. The formula calculates the interpolated water saturation ($S_{w,interpolated}$) at any intermediate

depth by assuming a linear gradient between the two known points. Once the simulation saturation is interpolated to the same depth levels as the well log, the saturation mismatch can then be calculated ($S_{wWell\ Log} - S_{wSimulation}$). This difference serves as the objective function for optimization, where model parameters such as capillary pressure endpoints and oil-water contact are adjusted to minimize the mismatch. Accurate alignment between simulation and log-derived saturations is critical for improving the quality of dynamic model initialization.

Uncertainty study

In this study, COSTA models that have 3, 5, and 7 rock types are used in the uncertainty study to generate data for machine learning proxy models. The number of uncertain parameters is directly tied to the number of rock types, as each rock type is characterized by its own capillary pressure – water saturation function. Given that each function is defined by three key endpoints (Pc Max, Swc, Sw Max) then a model that has 5 rock types, will require the calibration of 15 endpoints, in addition to one oil-water contact. For each rock typing scenario, 100 simulation cases were generated by systematically varying these parameters. The OWC was varied within a range of ± 50 ft from the original base value to reflect geological uncertainty, while the capillary pressure – water saturation endpoints were varied within $\pm 40\%$ of their original values to capture realistic ranges in petrophysical properties as shown in Table 1. It is important to note that the maximum value of Swc was not varied by the full 40% range to ensure it remained below the minimum value of Sw Max, and similarly, the minimum value of Sw Max was constrained to remain above the maximum Swc to preserve physical consistency.

Table 1—Uncertainty parameters of a COSTA model with 5 rock types. OWC varied ± 50 ft; capillary pressure–saturation endpoints varied $\pm 40\%$ to reflect geological and petrophysical uncertainty.

Parameters	Base	Min	Max
\$OWC	-8180	-8230	-8130
\$PC1	40	24	52
\$PC2	13.41	8.046	17.433
\$PC3	40	24	52
\$PC4	40	24	52
\$PC5	40	24	52
\$SWc1	0.84	0.504	0.88
\$SWc2	0.4	0.24	0.47
\$SWc3	0.29	0.174	0.34
\$SWc4	0.17	0.102	0.2
\$SWc5	0.08	0.048	0.1
\$SWmax1	0.95	0.93	0.96
\$SWmax2	0.9	0.54	0.96
\$SWmax3	0.93	0.558	0.96
\$SWmax4	0.93	0.558	0.96
\$SWmax5	0.7	0.42	0.82

After completing the uncertainty study, the resulting simulation data are organized into a structured tabular format, where each row represents a unique combination of input parameters and the corresponding saturation mismatch outcome. This dataset is then used to train a machine learning model, enabling it to

learn the underlying relationships and accurately predict the saturation mismatch based on variations in capillary pressure endpoints, and oil-water contact depth.

Machine Learning Proxy Model

In this study, machine learning methods will be used to build a proxy model to predict the saturation mismatch of the simulation based on the uncertain parameters. The idea of having a proxy model is to capture the uncertainty of the dynamic parameters when solving the capillary gravity equilibration in initialization. This proxy model will be then used to optimize the uncertain parameters to find the best configuration that can reduce the saturation mismatch between simulation and well log measurements.

A proxy model does not have to be built using machine learning techniques; it can also be developed using conventional methods such as polynomial fitting, which approximates the relationship between input parameters and output responses through a predefined mathematical equation (Suwito et al., 2022). However, this approach may be limited in capturing complex nonlinear interactions and may lose accuracy when applied to high-dimensional or highly variable datasets. On the other hand, machine learning models are highly effective at capturing complex patterns within large datasets and making accurate predictions due to their rapid learning capabilities.

Two predictive approaches were evaluated for predicting the water saturation mismatch. Method 1 involves predicting the average saturation mismatch per simulation case. After computing the mismatch across all depth points, the values are averaged across wells, resulting in a simplified dataset. For example, with 600 simulation cases and 16 input features, the resulting dataset shape is [16, 600], which significantly reduces computational cost and allows for faster machine learning training and optimization. Method 2 retains the full-resolution saturation mismatch without averaging. It accounts for well and depth-specific variations by predicting the mismatch at each depth point. In the COSTA model, which includes 15 well logs with 200 depth points each, and 600 simulation cases, this approach yields a much larger dataset of shape [16, 1,800,000]. While this increases the computational load, it preserves spatial heterogeneity and better captures the influence of local rock types, offering potentially improved predictive accuracy.

Several machine learning algorithms were tested to identify the most suitable model for predicting saturation mismatch, and random forest regression demonstrated the best performance in terms of accuracy and robustness. Random forest is an ensemble learning method that builds multiple decision trees during training as shown in Figure 4 and aggregates their predictions to improve generalization and reduce overfitting. This model is particularly effective for handling non-linear relationships and interactions among input features, making it well-suited for complex reservoir simulation problems. Once the random forest model was trained using the uncertainty study dataset, it was used to predict saturation mismatch and guide the optimization of capillary pressure endpoints and oil-water contact.

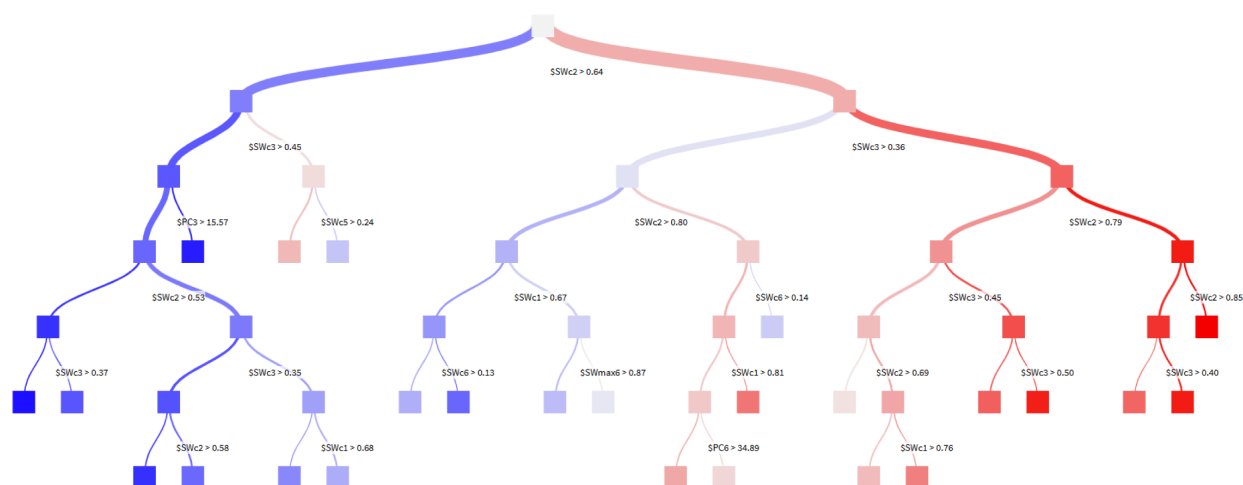


Figure 4—An Illustration of Random Forest Decision Tree to predict the saturation mismatch based on capillary pressure end points and OWC.

Optimization to Minimize the Saturation Mismatch

After the proxy model is trained, it is used as an objective function for optimization. In Method 2, where saturation prediction is performed at each depth level of the well, certain variables such as Well and Depth are held constant. The optimization process focuses only on adjusting the endpoints of the capillary pressure–saturation function and the OWC. For this task, the Powell optimization method was selected. Powell's method is an optimization algorithm that identifies local minima by performing sequential line searches along directions that are continually adjusted to improve efficiency (Powell, 1964). Compared to other optimization algorithms, Powell's method is computationally efficient and produces results more quickly, which is beneficial in iterative workflows in this study.

Results and Discussion

The dynamic calibration workflow was applied to three COSTA models configured with 3, 5, and 7 rock types. Each model was analyzed by developing its own machine learning proxy and optimization routine to evaluate improvements in saturation mismatch.

Saturation Mismatch Prediction

A total of 200 simulation cases were generated for each rock typing scenario to evaluate the effectiveness of machine learning proxy models in capturing saturation mismatch patterns. Two predictive approaches were implemented. Method 1 involved averaging the saturation mismatch across all depth points and wells for each simulation case. This significantly reduced data dimensionality, allowing for faster model training while still achieving high predictive accuracy. The random forest regression model attained R^2 values of 0.936, 0.972, and 0.933 for the three, five, and seven rock type scenarios, respectively.

Method 2 retained vertical resolution by predicting saturation mismatch at each depth point individually, thereby preserving the spatial heterogeneity of the reservoir. Although this method required significantly more data and longer training time, it allowed the model to capture more detailed variations across depth. The random forest regressor maintained strong predictive performance, with R^2 values of 0.921, 0.933, and 0.911 for the three, five, and seven rock type configurations, respectively. The performance of the model for the 7-rock type case is visualized in Figure 5, which shows the scatter plot of predicted versus actual saturation mismatch. The close clustering around the diagonal line indicates that the model reliably captured the uncertainty of dynamic parameters in predicting the saturation mismatch.

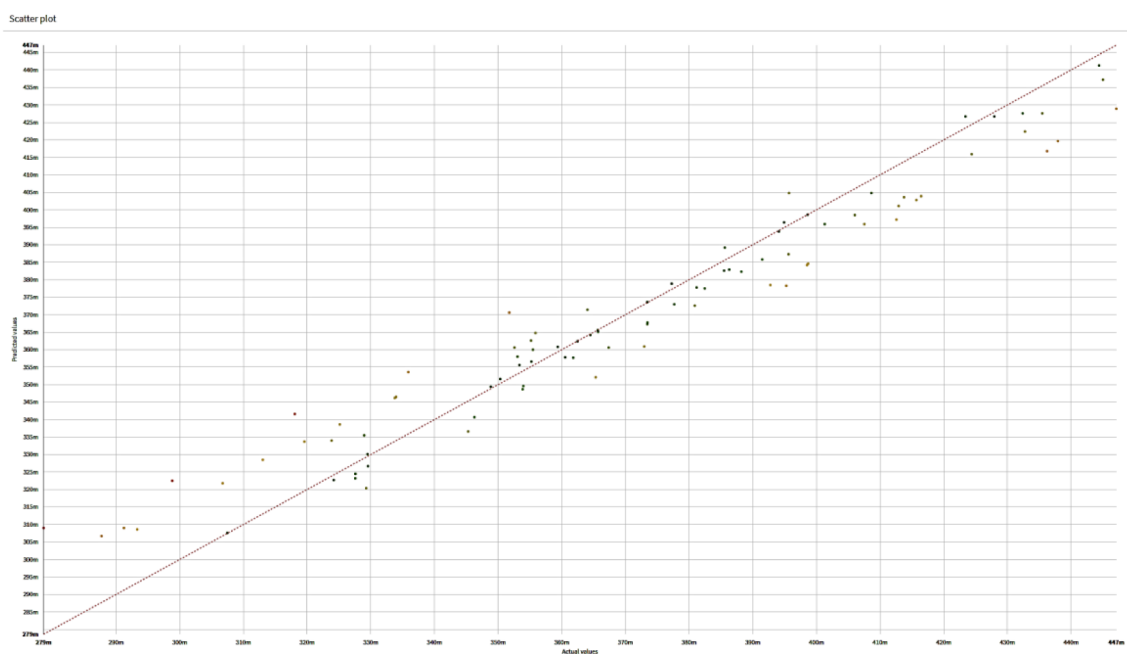


Figure 5—Scatter plot showing the distribution of actual versus predicted saturation mismatch using the random forest regressor for Method 2 applied to the 7-rock type model.

Saturation Initialization Optimization

After building the machine learning proxy model to predict the saturation mismatch, each machine learning model was coupled with an optimization model aimed at minimizing the predicted saturation mismatch by adjusting the capillary pressure endpoints and OWC. This process systematically searched the parameter space to identify the combination that yielded the lowest mismatch between simulated and log-derived water saturation. The optimization process was applied to both predictive approaches of Method 1 and Method 2, for each of the three rock type configurations.

Figure 6 illustrates the resulting saturation mismatch that is averaged for each well, before and after calibration for each rock typing scenario. In the original COSTA model, the average saturation mismatches for all 15 wells were 0.28, 0.34, and 0.33 for models with 3, 5, and 7 rock types, respectively. Using Method 1, the optimized models reduced these mismatches to 0.21, 0.22, and 0.25, respectively. In contrast, Method 2, which preserves vertical heterogeneity by predicting mismatch at each depth point, achieved even greater improvements, reducing the mismatches to 0.20, 0.17, and 0.19. Among all cases, the model configured with 5 rock types demonstrated the most substantial reduction, especially with the prediction Method 2.

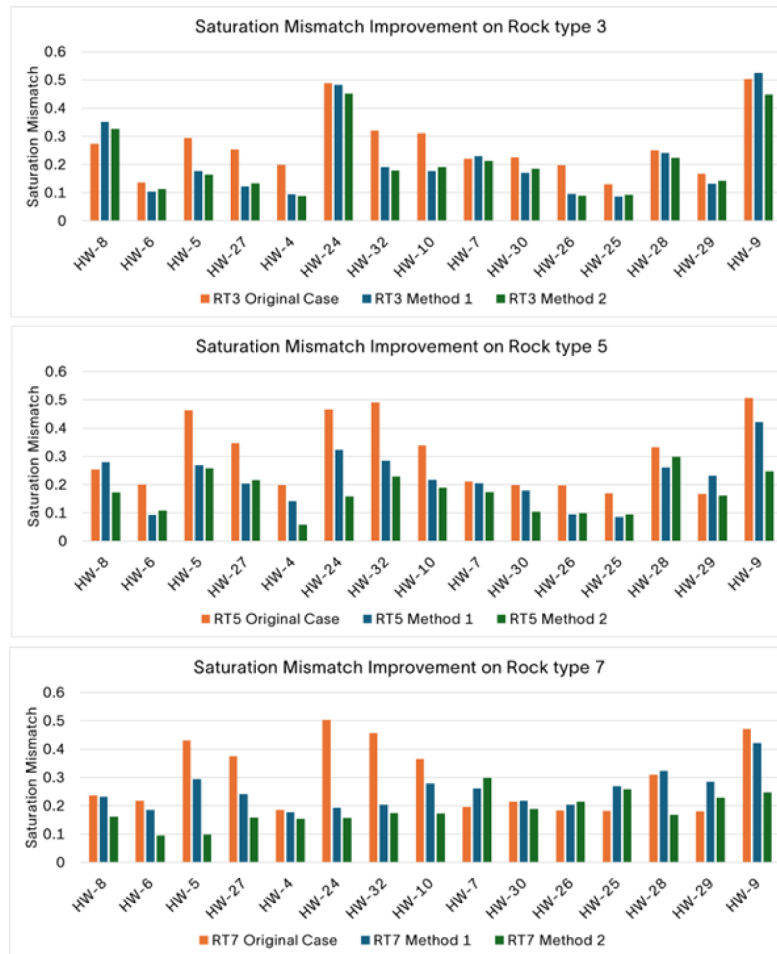


Figure 6—Comparison of saturation mismatch before and after calibration for both Method 1 and Method 2, (a) Top chart shows comparison on the case with Rock type 3; (b) Middle chart shows comparison on the case with Rock type 5; and (c) Bottom chart shows comparison on the case with Rock type 7.

Figure 6 demonstrates that Rock Type 5, with Method 2, yields the most substantial reduction in saturation mismatch across most wells. These improvements are further detailed in Figure 7, which presents well-level comparisons of water saturation before and after calibration of the capillary pressure endpoints and oil–water contact OWC. In Figure 7, the black line represents water saturation derived from well log measurements, the red line corresponds to the water saturation from reservoir simulation initialization of the original COSTA model with 5 rock types, and the blue line indicates the water saturation after the dynamic calibration with Method 2. It is evident that, following calibration, the initialization of water saturation more closely aligns with the well log measurements, confirming the effectiveness of the optimization in enhancing the match between static and dynamic data. These results confirm that integrating machine learning with parameter optimization can significantly improve saturation initialization in complex carbonate reservoirs.

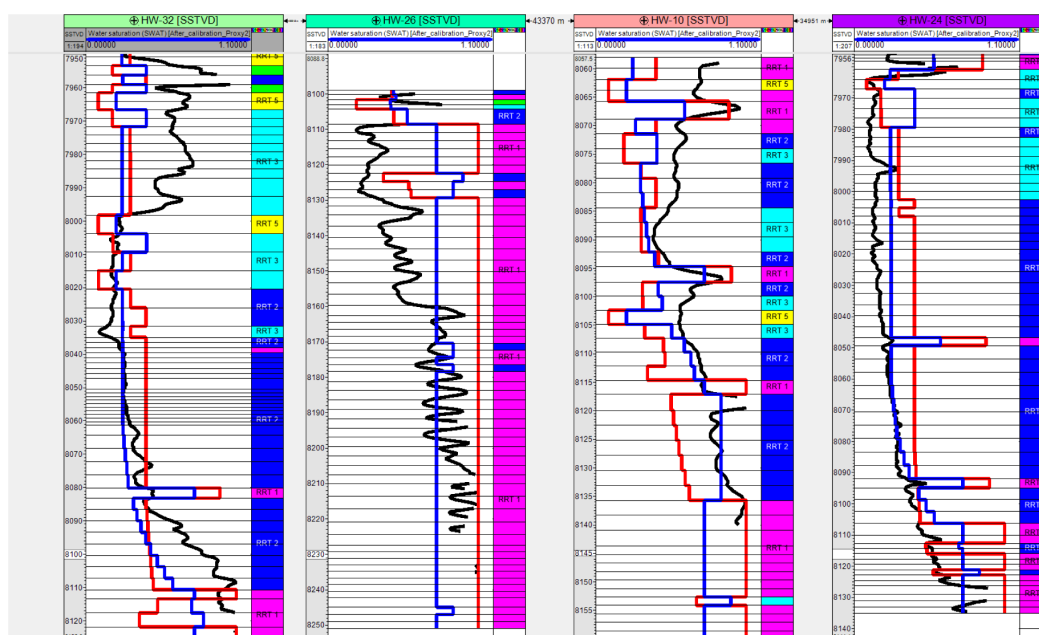


Figure 7—Well-level water saturation comparisons for wells HW-32, HW-26, HW-10, and HW-24, illustrating the impact of calibrating uncertain parameters. The plots show water saturation before calibration (red line) and after calibration (blue line) against well log measurements data (black line).

Numerical stability after calibration

As the water saturation from reservoir simulation initialization becomes closer to the well log measurements, the model more accurately reflects the actual reservoir conditions. This improved alignment is expected to enhance computational accuracy and reduce convergence issues. The calibrated saturation initialization not only provides a better match with well log data but also contributes to improved simulation performance. As shown in Figure 8b, the number of Newton iterations required after calibration is generally lower than in the original case, indicating more stable numerical convergence. This improvement leads to faster simulation runtimes, as illustrated in Figure 8a, where the total runtime is reduced by approximately 52% compared to the original case, reducing the simulation time from 29.5 minutes to 14 minutes. Furthermore, the more accurate initial conditions are expected to support a faster and more reliable history matching process. Overall, this study demonstrates the value of integrating machine learning and optimization into the initialization workflow to reduce uncertainty, improve model representativeness, and enhance the overall efficiency of reservoir simulation.

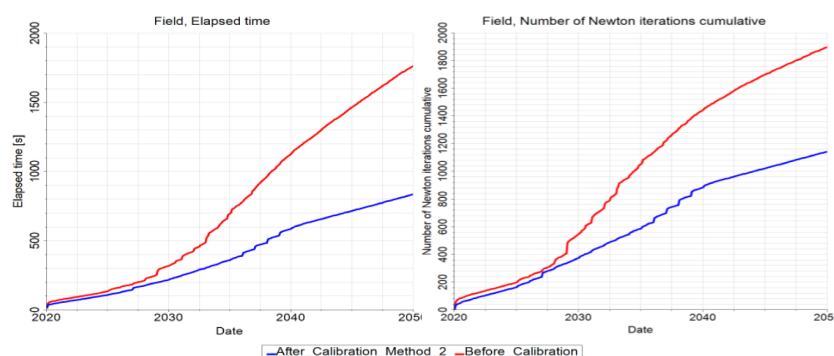


Figure 8—Simulation performance comparison over 30 years of reservoir simulation for the COSTA model with 5 Rock Types, (a) The left chart shows the total simulation runtime; (b) The right chart illustrates the cumulative number of Newton iterations, highlighting improvements in numerical stability after calibration.

Conclusion

This study introduces an integrated workflow that combines machine learning with dynamic parameter optimization to calibrate reservoir simulation initialization more efficiently and accurately. Three configurations of the COSTA model, featuring different numbers of rock types, were used to evaluate the impact of geological heterogeneity on water saturation initialization. The calibration process began with the calculation of saturation mismatch between simulation-derived and well log-derived saturation values. A machine learning proxy model was then developed using 200 realizations for each rock type generated from an uncertainty analysis, where the capillary pressure–saturation endpoints and oil–water contact OWC were varied.

Two calibration approaches were investigated: Method 1, which predicts and optimizes the average saturation mismatch per simulation case, and Method 2, which predicts and optimizes saturation mismatch at each depth point. Method 2 provided superior results by preserving vertical heterogeneity and allowing for more precise optimization. Among the tested models, the configuration with five rock types yielded the most favorable outcome, reducing the average saturation mismatch from 0.34 to 0.17. This improvement also led to more stable numerical performance, as indicated by a reduction in the number of Newton iterations and a 52% decrease in the simulation runtime.

The main limitation of the workflow presented lies in its dependence on the initial engineering design of the rock type model and capillary pressure function. If the rock typing or capillary pressure characterization is poorly defined, the calibration improvements may be limited in their ability to accurately match water saturation. On the other hand, this limitation can serve as a valuable validation tool, highlighting which rock type scenario is more representative and guiding a better engineering approach.

Furthermore, the current workflow focuses primarily on calibrating the end points of the capillary pressure function. This parameterization can also be extended to a J-function–based approach, which represents a natural direction for future work. In addition, since Method 2 introduces higher computational demand, exploring more efficient optimization strategies will be an important step toward improving the scalability and practical implementation of the workflow.

Overall, the proposed workflow demonstrates a systematic and automated approach to dynamic calibration, offering a more effective alternative to manual trial-and-error methods. It enhances model reliability, accelerates simulation processes, and provides a robust framework for determining an appropriate number of rock types. This contributes to better uncertainty quantification and supports more informed decision-making in reservoir development planning.

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